



Department of Information Technology/Computer Science  
Final year Individual Research Project  
CS7019/IT8029/IS8029

## Project Norm Form

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### Student Information

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### Title of the Project:

Automating Waste Collection with Smart Technology for Optimized Operations.

### Problem in Brief:

Inefficient waste collection systems in urban areas lead to environmental pollution, public health hazards, and increased operational costs. Overflowing bins, delayed pickups, and poorly planned routes exacerbate these issues, causing inconvenience to both residents and waste collection authorities. These problems are often a result of existing systems that rely on manual monitoring and reporting, which are time-consuming, unreliable, and lack real-time insights. This gap underscores the urgent need for a smart, automated solution capable of tracking waste bins and collecting trucks in real time, optimizing collection routes, notifying users promptly, and enabling efficient communication between service providers and citizens.

### Solution:

The system optimizes waste collection by automatically detecting bin fill levels through IoT-enabled sensors and sending real-time updates to relevant authorities. It ensures efficient waste collection by notifying collection teams and users about bin statuses and scheduled pickups. Additionally, the system provides real-time tracking of waste collection trucks to ensure timely operations and minimize delays.

Through mobile and web applications, the platform allows users to receive instant notifications, track collection updates, and request pickups when necessary. By automating monitoring and communication, the system enhances waste collection efficiency, reduces operational inefficiencies, and promotes a cleaner urban environment.

**Technologies:**

- ASP.NET (Backend)
- Flutter (Mobile App)
- 03.React (Web Application)
- 04.Firebase (Database)
- 05.Nordic MCU (Microcontroller – IoT Bin)
- 06.Google Maps API (Location and Mapping Services)

**Area of the Research Project**

A Comprehensive Review of Modern Technologies in Waste Collection: Driving Smarter and Greener Solutions for a Future-Ready Approach.

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**FOR OFFICE USE**

**Research Group:**

**Supervisor(s)\*:**